Name __________________________ **Class: 16S**_______ **Reg Number:** _____

MERIDIAN JUNIOR COLLEGE
JC 2 Preliminary Examination
Higher 2

Chemistry **9729/03**

Paper 3 Free Response **13 September 2017**

2 hours

Additional Materials: *Data Booklet*
Writing Paper

INSTRUCTIONS TO CANDIDATES

Write your name, class and register number in the spaces provided at the top of this page.

Answer all questions in Section A and one question from Section B.

Begin each question on a fresh page of writing paper.

Fasten the writing papers behind the given **Cover Page for Questions 1 & 2** and **Cover Page for Questions 3 & 4 or 5** respectively.

Hand in Questions **1 & 2** and **3 & 4 or 5** **separately**.

You are advised to spend about 30 minutes per question only.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

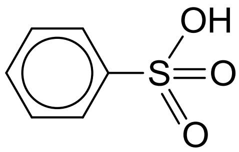
This document consists of **20** printed pages.

Section A

Answer **all** questions in this section.

1 Sulfa drugs are a class of sulfur containing compounds. These drugs have a variety of uses and can be classified as antibiotics and non–antibiotic drugs. They were the "wonder drugs" before penicillin was discovered.

A precursor to synthesising such sulfa drugs is aryl sulfonic acid with the general structure shown below.



Aryl sulfuric acids may be synthesised from the electrophilic substitution reaction between concentrated sulfuric acid and an aromatic compound.

(a) In an experiment to investigate the rate of reaction between concentrated sulfuric acid and benzene to form sulfonic acid, different volumes of sulfuric acid and benzene were used and the time taken for a complete reaction was recorded in **Table 1.1**.

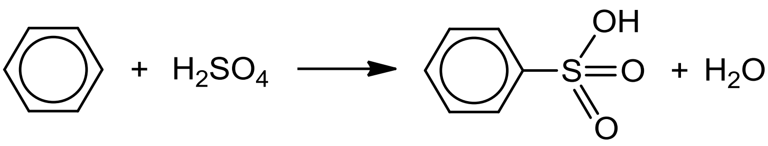
Table 1.1

Experiment	**Volume of concentrated sulfuric acid / cm^3**	**Volume of benzene / cm^3**	**Volume of water / cm^3**	**Time taken for complete reaction / s**
1	10	20	30	60
2	20	20	20	15
3	20	30	10	10

(i) Use the data above to deduce the order of reaction with respect to concentrated sulfuric acid and benzene, showing how you arrive at your answer. [3]

(ii) Hence, construct the rate equation for the reaction. [1]

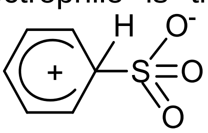
(b) The overall equation for the formation of the aryl sulfonic acid, $C_6H_5SO_3H$, through electrophilic substitution is given below.

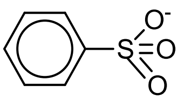


This reaction proceeds via a four–step mechanism.

- **Step 1**: The SO_3 electrophile is first generated from concentrated sulfuric acid given by the following equation.

  $$2H_2SO_4 \rightleftharpoons SO_3 + H_3O^+ + HSO_4^-$$

- **Step 2**: The SO_3 electrophile is then attacked by benzene to form an intermediate  .

- **Step 3**: When the intermediate from **Step 2** is deprotonated, an anion  is formed and the sulfuric acid is regenerated.

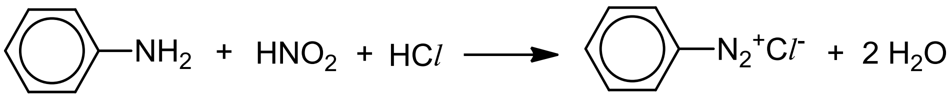
- **Step 4**: The $C_6H_5SO_3^-$ anion is then protonated by a hydronium ion, H_3O^+, to form the products, $C_6H_5SO_3H$ and water.

(i) Use the information given above to outline the mechanism for **Steps 2 to 4**, showing all charges and using curly arrows to show the movement of electron pairs. You are required to show the displayed formulae of all species. [3]

(ii) With reference to your answer in **(a)(ii)**, suggest which step corresponds to the rate determining step of the mechanism. [1]

(c) Another application of aryl sulfonic acid is in coupling with diazonium salts to form useful compounds.

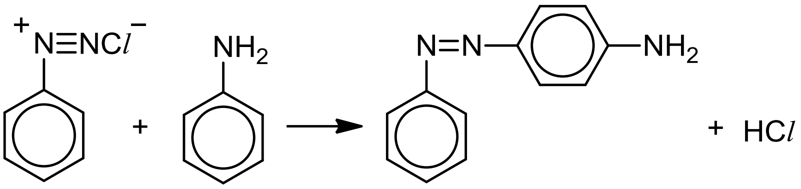
The diazonium salt is first produced by reacting phenylamine with cold nitrous acid and hydrochloric acid. This process is called diazotisation.



diazonium salt

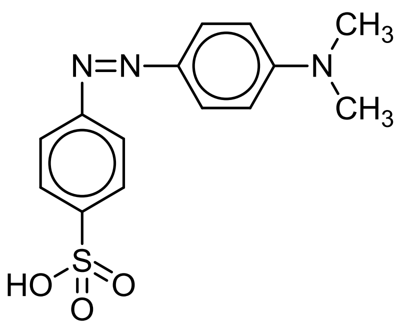
(i) Describe how you will carry out a simple chemical test to distinguish between phenylamine and the diazonium salt. [2]

The diazonium salt obtained from the diazotisation can undergo coupling with another phenylamine to form an azo compound, which is a useful dye. An example of a coupling reaction is given below.



diazonium salt azo compound

Methyl orange, which is used as an acid–base indicator, is formed using coupling reaction between a phenylamine derivative and a sulfonic acid containing diazonium salt.



methyl orange

(ii) Draw the structures of the two reactants used in the coupling reaction to synthesise methyl orange. State which of these two reactants is the electrophile. [3]

(d) Similar to sulfuric acid, concentrated sulfonic acids can be used as a catalyst in the synthesis of esters.

Compound **P**, $C_7H_{13}O_2Br$, is formed from the reaction between compounds **Q** and **R**, using concentrated sulfonic acid as a catalyst. Both compounds **Q** and **R** are able to rotate plane–polarised light.

Upon combustion with excess oxygen, 0.1 mole of compound **Q** produces 0.5 mole of carbon dioxide gas and 0.5 mole of water. Effervescence is observed when two moles of compound **Q** reacts with one mole of sodium carbonate.

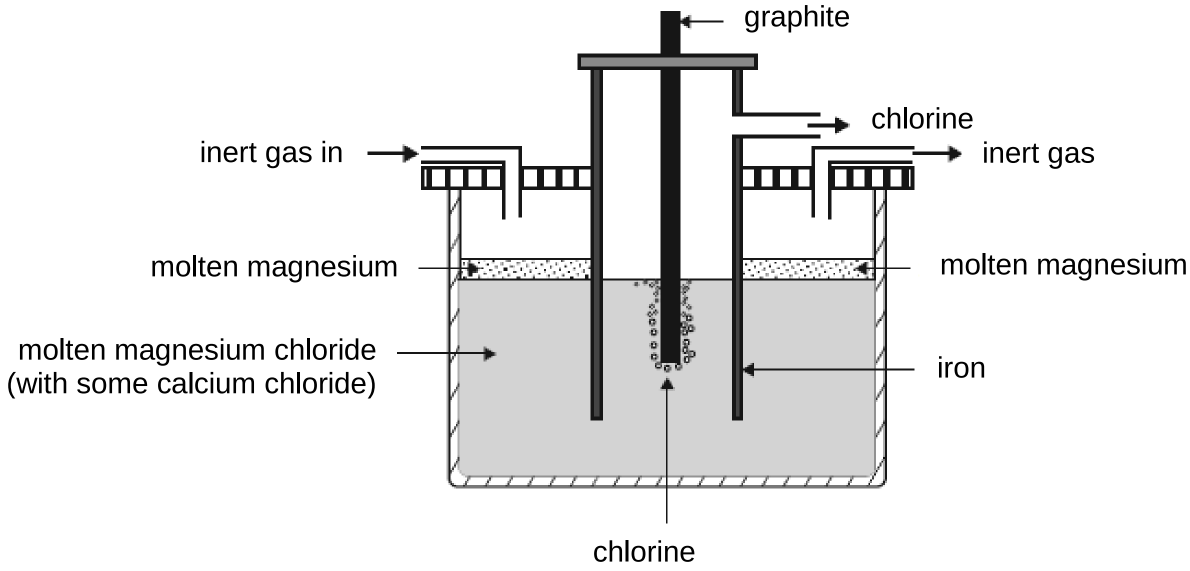
Compound **R** gives a yellow precipitate when heated in an alkaline solution of iodine. Upon heating with sodium hydroxide, followed by the addition of aqueous silver nitrate, a cream precipitate was formed.

Deduce the structures of **P**, **Q** and **R**, explaining the chemistry of the reactions described. [7]

[Total: 20]

2 Magnesium–containing compounds are important substances that have found many applications in our daily life.

(a) Magnesium metal is one of the most abundant elements on Earth. It is produced by the electrolysis of molten magnesium chloride. An example of such an electrolytic cell to form molten magnesium is shown below.



(i) Write the half equations, with state symbols, for the reactions at the graphite and iron electrodes. [2]

(ii) Explain why an inert gas is constantly blown into the compartment. [1]

(iii) Generally, to reduce energy consumption, $CaCl_2$ and $ZnCl_2$ can be added to lower the melting point of $MgCl_2$. However, in this electrolytic cell, only $CaCl_2$ can be used for this purpose.

With reference to $E^{\ominus}$ values in the *Data Booklet*, explain why only $CaCl_2$ can be used to lower the melting point of $MgCl_2$ in this electrolytic cell and not $ZnCl_2$. [2]

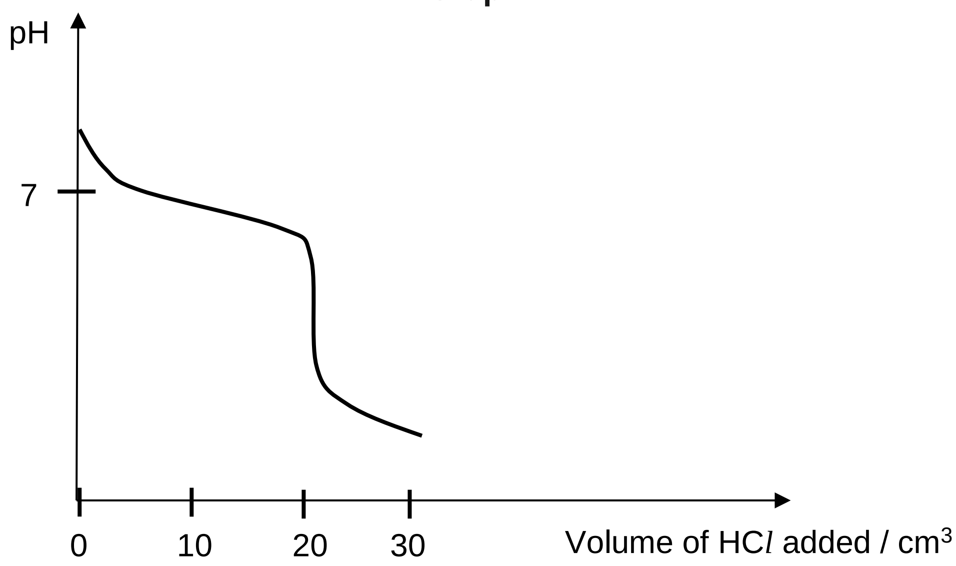
(iv) By quoting relevant $E^{\ominus}$ values, explain whether an iron electrode is a suitable replacement for the graphite electrode for the above electrolytic cell. [2]

(v) Another metal, **M**, is obtained by using the same electrolytic cell above. During the electrolysis, a current of 1.5 A was applied for 6 hours. The amount of metal **M** obtained was 1.12×10^{-1} mole. Determine the charge of the molten metal ion, $\mathbf{M}^{x+}$. [2]

(b) Magnesium hydrogencarbonate, $Mg(HCO_3)_2$, is a bicarbonate salt of magnesium with $K_b = 2.4 \times 10^{-8}$ mol dm^{-3}.

A sample of 0.10 mol dm^{-3} $Mg(HCO_3)_2$ solution was titrated against a 0.10 mol dm^{-3} solution of hydrochloric acid. The following titration curve was obtained.

Graph 2.1



(i) With the aid of the titration curve, calculate the volume of the sample of $Mg(HCO_3)_2$ solution. [1]

(ii) Calculate the initial pH on the titration curve. [2]

(iii) With the aid of an equation, explain why the pH value obtained at equivalence point is acidic. No calculations are required. [2]

(iv) Given the pH transition ranges of some indicators in **Table 2.1**, suggest with reasoning, an appropriate indicator for the titration between aqueous $Mg(HCO_3)_2$ and hydrochloric acid.

Table 2.1

Name of indicator	pH transition range
bromocresol green	4 – 6
thymolphthalein	9 – 11

[2]

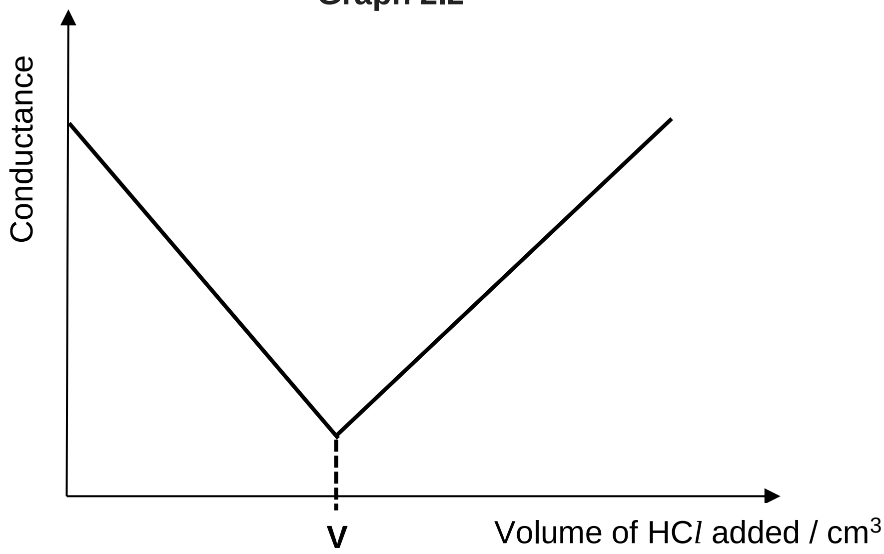
(v) State the volume of hydrochloric acid and calculate the pH at which the buffer operates at its maximum buffer capacity. [2]

(vi) As compared to magnesium hydrogencarbonate, magnesium carbonate has a K_b value of 2.10×10^{-4}. Explain why magnesium carbonate is a stronger base than magnesium hydrogencarbonate. [1]

(c) Instead of monitoring the pH change during the titration, the electrolytic conductivity of the reaction mixture is continuously monitored. The equivalence point is the point at which conductivity undergoes a sudden change.

The conductometric titration curve shown in **Graph 2.2** is obtained when 20 cm^3 of 0.10 mol dm^{-3} aqueous sodium hydroxide was titrated against 0.10 mol dm^{-3} aqueous hydrochloric acid. The change in conductance may be assumed to be only due to the changes in the amount of hydrogen ions and hydroxide ions.

Graph 2.2



Using your understanding of the titration between sodium hydroxide and hydrochloric acid, determine the volume **V** in the above conductometric titration curve. Explain the change in conductance as an increasing volume of HC*l* is added.

[3]

[Total: 22]

3 This question is about the chemistry of oxygen containing compounds.

(a) "Elephant's toothpaste" is a foamy substance caused by the rapid decomposition of hydrogen peroxide as shown by the equation:

$$2H_2O_2 \rightarrow 2H_2O + O_2$$

(i) The decomposition of H_2O_2 results in the formation of a highly reactive species, •OH, with an unpaired electron. With the aid of an equation and the use of curly arrows to represent the movement of electrons, show how this reactive species may be formed from H_2O_2. [1]

(ii) Using data from the *Data Booklet*, calculate $E^{\ominus}_{cell}$ for the decomposition of hydrogen peroxide. [1]

(iii) Hence, calculate $\Delta G^{\ominus}$ for the decomposition of hydrogen peroxide. Explain the significance of your answer. [2]

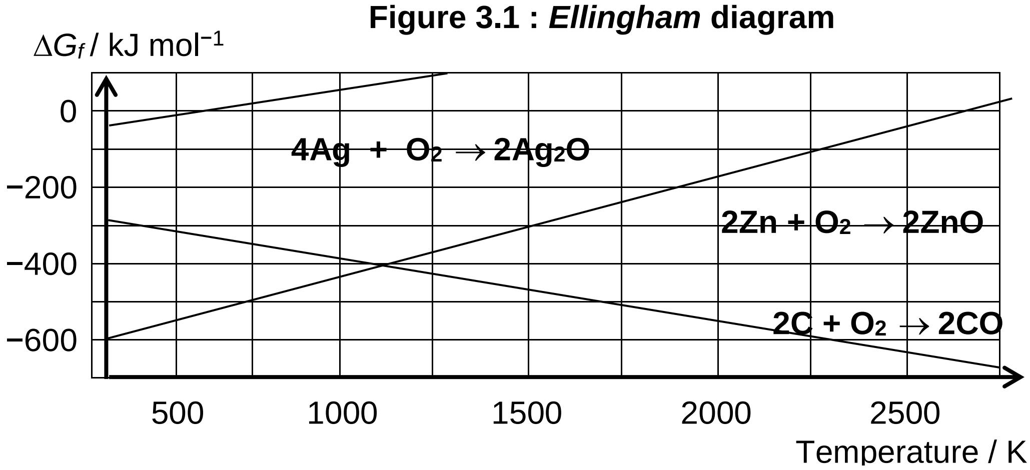
(iv) Fe^{2+}(aq) ions can be used to catalyse the decomposition of hydrogen peroxide. Using relevant data from the *Data Booklet*, describe and explain the role of Fe^{2+}(aq) ions in this reaction. You should support your answers with relevant equations. [3]

(b) ΔG, ΔH and ΔS are related by the following equation.

$$\Delta G = \Delta H - T\Delta S$$

Figure 3.1 is an *Ellingham* diagram, which shows the variation in the Gibbs free energy change of **formation**, ΔG_f, with temperature, T, for some oxides.

Figure 3.1 : *Ellingham* diagram



(i) With reference to the gradient of the graph for the formation of zinc oxide, ZnO, explain with reasoning, the sign of ΔS. [2]

(ii) The *Ellingham* diagram in **Figure 3.1** can also be used to deduce the ease of decomposition of silver oxide and zinc oxide. Suggest which metal oxide is thermally less stable at 1000 K. Explain your answer. [2]

(iii) Similar to enthalpy change of reaction, the Gibbs free energy, ΔG, of a reaction can be calculated using the following expression.

$$\Delta G = \sum \Delta G_f(\text{products}) - \sum \Delta G_f(\text{reactants})$$

Calculate the ΔG value for the reduction of zinc oxide by carbon at 2250 K. Hence, comment on the feasibility of this reaction at 2250 K. [2]

(c) The zinc–air battery is commonly used as small button cells in watches and hearing aids. When a gas permeable and liquid–tight membrane sealing tab in the button cell is removed, oxygen in the air is absorbed into an alkaline electrolyte.

The positive electrode is made of porous carbon and the negative electrode consists of zinc. The zinc electrode slowly dissolves to form a colourless solution. The electrolyte used is a paste of potassium hydroxide.

(i) Write equations, with state symbols, for the reactions that occur at the anode and the cathode respectively. [2]

(ii) The zinc–air battery has a cell potential of +1.59 V. Using relevant data from the *Data Booklet*, calculate the electrode potential for the reaction at the anode. [1]

(iii) The zinc electrode of a new zinc–air button cell weighs 1.85 g. The cell can run until 80% of the zinc is consumed.

Calculate the maximum amount of current that can be drawn from the cell if it is expected to last for 365 days. [1]

(iv) Common alkaline batteries contain zinc and manganese(IV) oxide in a paste of potassium hydroxide. The overall equation is as shown.

$$Zn\ (s) + MnO_2\ (s) + H_2O\ (l) \rightarrow ZnO\ (s) + Mn(OH)_2\ (s)$$

Based on this information, suggest an advantage, other than lower cost that the zinc–air battery has over the common alkaline battery of a similar mass. Explain your answer. [1]

[Total: 18]

Section B

Answer **one** question from this section.

4 Barium hydroxide is a compound that can exist in the anhydrous or hydrated forms. Its main uses are as a precursor for the manufacture of other barium compounds and as a substance to remove sulfates from various products.

(a) **(i)** Draw the dot–and–cross diagram for sulfate(VI) ion. In your diagram, use the symbol ‘x’ and ‘•’ to distinguish between the electrons from sulfur and oxygen, and the symbol ‘Δ’ for any additional electrons responsible for the overall negative charge. [1]

(ii) Using the *Valence Shell Electron Pair Repulsion* theory, explain the shape of sulfate(VI) ion. [2]

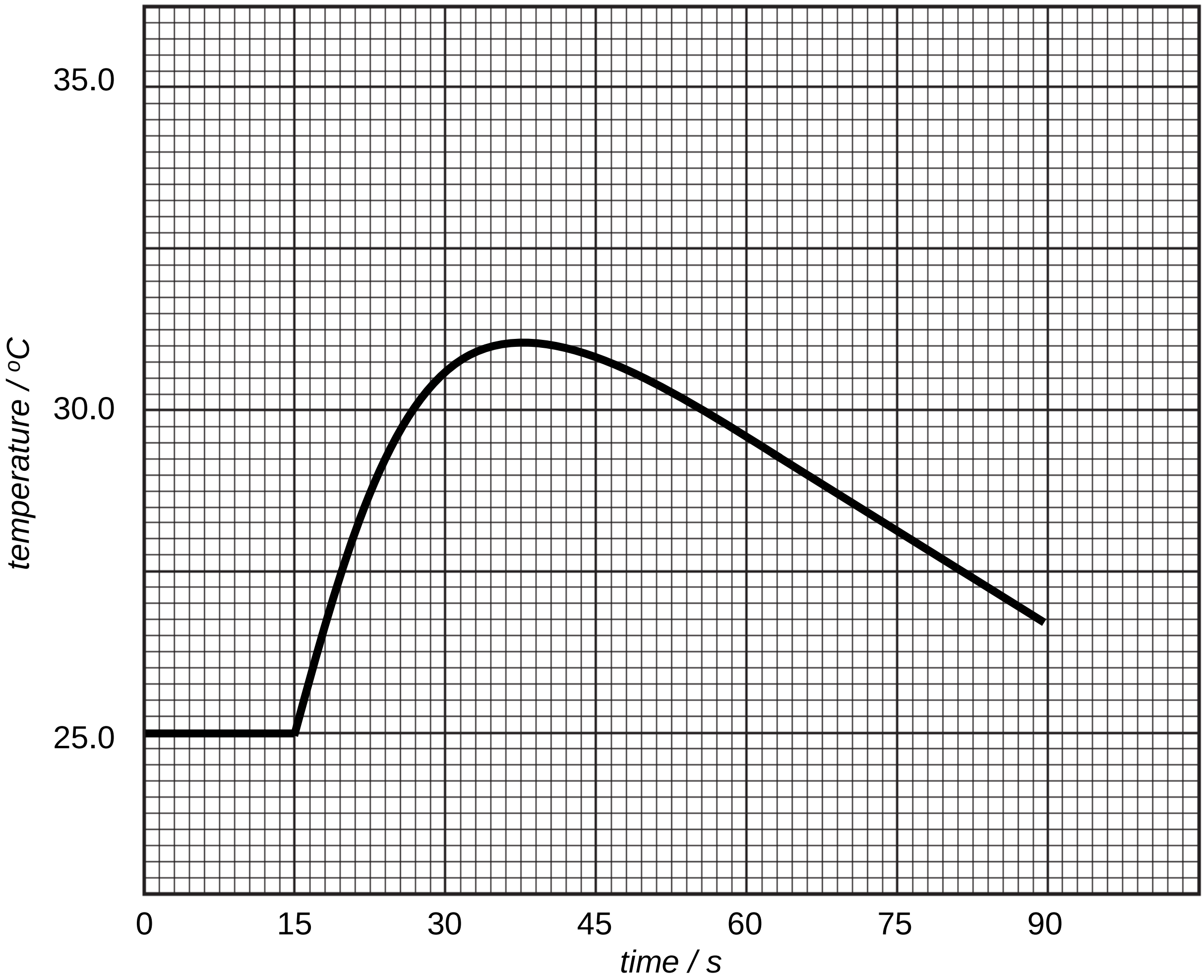
(b) Barium ion most commonly exists in the octahydrate form while magnesium ion usually exists in the hexahydrate form. When a 3.00 g solid sample of $Ba(OH)_2.8H_2O$ is strongly heated, the experimental loss in mass is 1.54 g.

$$Ba(OH)_2.8H_2O\ (s) \rightarrow BaO\ (s) + 9H_2O\ (g)$$

(i) By means of calculations, explain whether the decomposition of the $Ba(OH)_2.8H_2O$ sample is complete. [2]

(ii) Explain whether the thermal decomposition temperature of the anhydrous form of barium hydroxide is higher or lower than that of anhydrous magnesium hydroxide. [2]

(c) A temperature probe is used to measure the temperature of 100 cm^3 of 0.8 mol dm^{-3} barium hydroxide in a polystyrene cup. At 15 seconds, 120 cm^3 of 2.0 mol dm^{-3} hydrochloric acid is added. The temperature readings are recorded using a data logger for 90 seconds. The results are shown below.



(i) Construct appropriate lines on the graph above and determine the temperature rise of the reaction. [1]

(ii) Using your answer in **(c)(i)**, calculate a value for the enthalpy change of neutralisation for the reaction. [2]

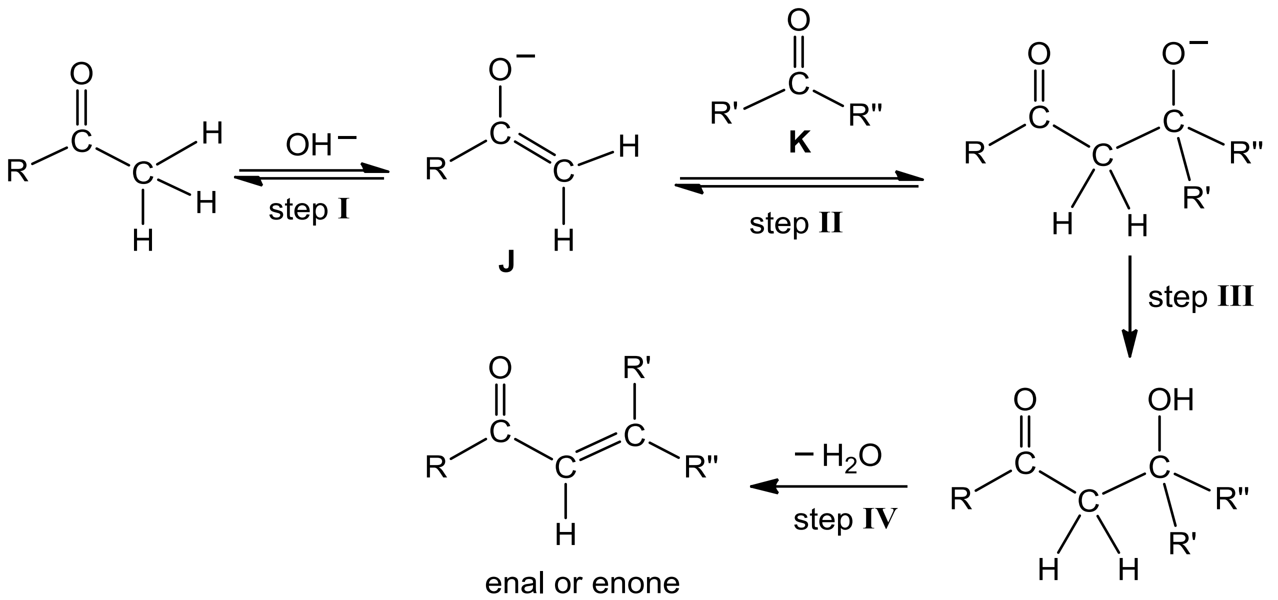
(d) Using the following data, calculate a value for the lattice energy of barium hydroxide by means of a *Born–Haber* cycle.

enthalpy change of atomisation of barium	= +182 kJ mol^{-1}
enthalpy change of formation of barium hydroxide	= –940 kJ mol^{-1}
$\frac{1}{2}O_2$ (g) + $\frac{1}{2}H_2$ (g) + $e^- \rightarrow OH^-$ (g)	= +230 kJ mol^{-1}

[3]

(e) Barium hydroxide may be used as a strong base for the first step of the *aldol condensation* reaction to produce an 'enal' or an 'enone'. An 'enal' is a molecule with an alkene group adjacent to an aldehyde group. An 'enone' is a molecule with an alkene group adjacent to a ketone group.

The following flowchart shows the *aldol condensation* reaction.

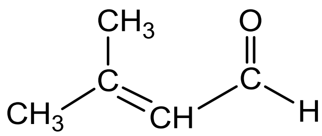


(R, R' and R'' = alkyl group or H atom)

(i) State the roles of species **J** and **K** in step **II**. [1]

(ii) State the type of reactions for steps **III** and **IV**. [1]

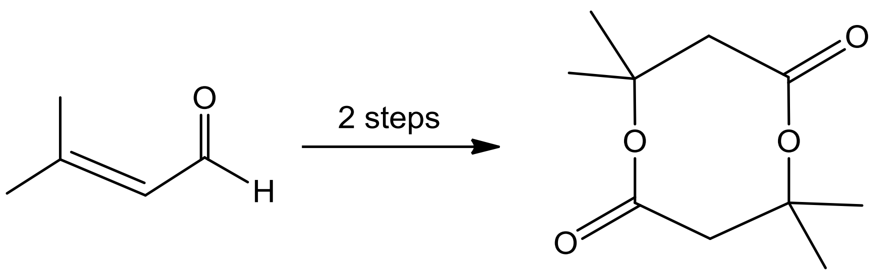
(iii) Suggest the structures of the two reactants required to synthesise the 'enal',



. [2]

(iv) In certain organic chemistry reactions, concentrated sulfuric acid and concentrated phosphoric acid may be used interchangeably.

Using the above information, propose a two–step synthesis route to convert the enal $(CH_3)_2C{=}CHCHO$, as the only available organic reactant, to compound **Q**.



Q

State the reagents and conditions required as well as the structures of the intermediates formed. [3]

[Total: 20]

5 Hydrogen cyanide, HCN, is extremely toxic and with sufficient concentrations it leads to rapid death. During the Second World War, a form of hydrogen cyanide known as *Zyklon B* was used in the Nazi gas chambers.

(a) Draw a dot–and–cross diagram to illustrate the bonding in HCN. [1]

(b) The synthesis of HCN was developed in the early 1900s. The most commonly used procedure is the *Andrussow* process. A less common method is the *BMA* process.

Andrussow process: $CH_4\,(g) + NH_3\,(g) + \frac{3}{2}\,O_2\,(g) \rightarrow HCN\,(g) + 3\,H_2O\,(g)$

BMA process: $CH_4\,(g) + NH_3\,(g) \rightarrow HCN\,(g) + 3\,H_2\,(g)$

(i) Using information from the *Data Booklet*, calculate the enthalpy change of reaction for each of the two processes above. [2]

(ii) Hence, comment on why the *Andrussow* process is the preferred procedure. [1]

(c) Other than gaseous HCN, sodium cyanide and potassium cyanide are used in the powder form in many homicide incidents.

It was predicted that salts like zinc cyanide, $Zn(CN)_2$, are harmless. The solubility data of some of these cyanides are given below.

solubility of NaCN at 25 °C = 1.3 mol dm^{-3}
solubility of KCN at 25 °C = 1.1 mol dm^{-3}
K_{sp} of $Zn(CN)_2$ = 8.0×10^{-12} $mol^3\,dm^{-9}$

Using the above information, explain why it was predicted that $Zn(CN)_2$ is harmless. [2]

(d) When cyanide is transported to the body's tissues, it binds irreversibly to an enzyme called *cytochrome c oxidase* and stops cells from being able to use oxygen for respiration. *Cytochrome c oxidase*, which contains Fe^{2+}, converts oxygen to water during respiration.

The United States standard cyanide antidote kit was developed in the last decade as a first aid tool. It comprises of a three step process.

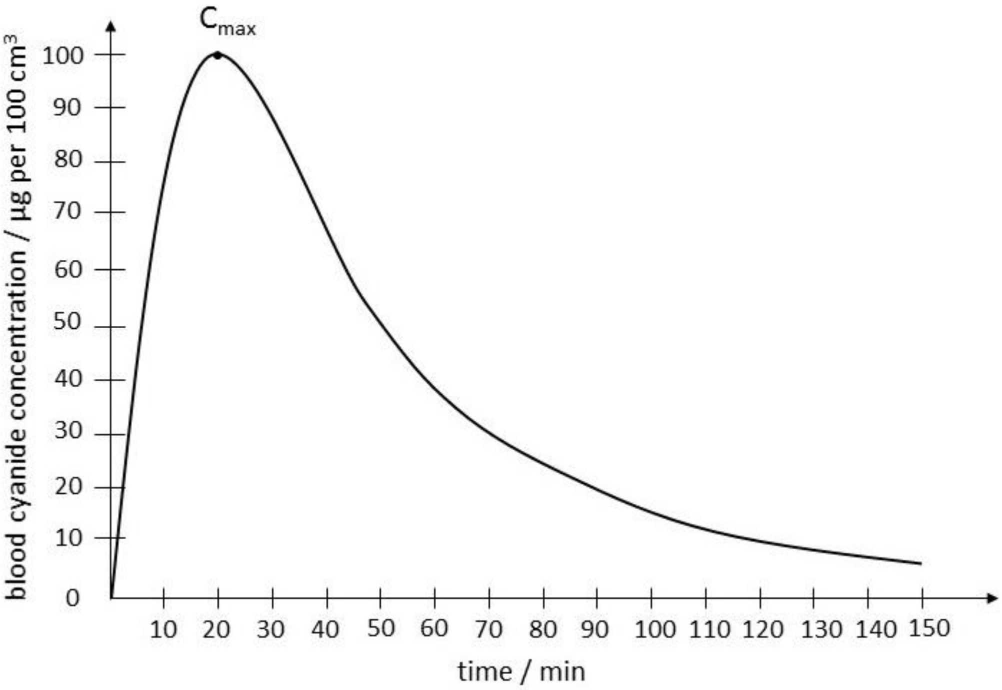
Step 1: Inhalation of a small dosage of amyl nitrite.
Step 2: Intravenous delivery of sodium nitrite, $NaNO_2$.
Step 3: Intravenous delivery of sodium thiosulfate.

(i) The nitrite oxidises some of the Fe^{2+} in haemoglobin in the red blood cells to Fe^{3+}, where Fe^{3+} has a higher affinity for cyanide. Suggest an advantage and a disadvantage of using nitrite as an antidote for cyanide poisoning. [2]

(ii) The sodium thiosulfate converts cyanide into thiocyanate ions, SCN^-, which is relatively harmless to the human body. Suggest, using coordination chemistry, how this conversion serves as an antidote. [1]

The human body has a mechanism which rapidly converts cyanide into the harmless thiocyanate via a first order kinetics when small doses are ingested. The cyanide concentration can be traced and represented by the **Graph 5.1**. **C_{max}** refers to the maximum concentration of cyanide in the body after ingestion. After which, it is eliminated by converting it to thiocyanate.

Graph 5.1



(iii) The safe level of cyanide concentration in blood is 20 µg per 100 cm^3 and below. From 20 to 200 µg per 100 cm^3, it is at a toxic level where the body may develop complications if the level is not brought back to safety after more than 60 minutes. (*Note: 1 µg = 1 x 10^{-6} g.*)

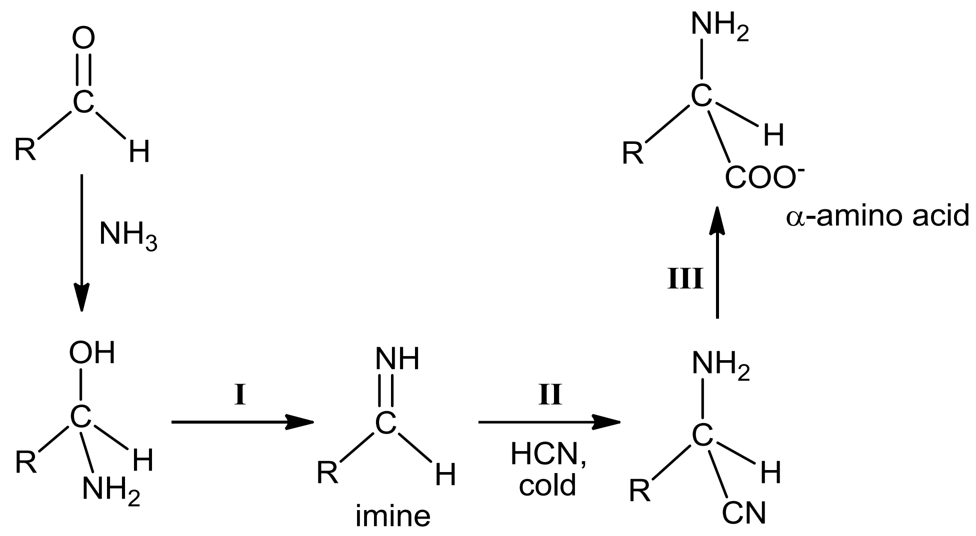
Using the graph and the above information, calculate the maximum cyanide blood concentration a victim can tolerate before complication occurs. [2]

In the octahedral complex, the d–subshell of the Fe^{3+} ion is split into two energy levels. In a 'high spin' state, the electrons occupy all the d–orbitals singly before pairing up in the lower energy d–orbitals. In a 'low spin' state, the lower energy d–orbitals are filled first, by pairing, before the higher energy d–orbitals are used.

(iv) Explain why splitting of the d–subshell occurs in the octahedral complex. [1]

(v) When cyanide binds with Fe^{3+}, it forms an octahedral complex which is at a 'low spin' state. Draw the electronic distribution of Fe^{3+} ion in its d–orbitals, showing the two energy levels. [1]

(e) Despite its lethal effect, hydrogen cyanide is often used in organic synthesis as a precursor to many important products like amino acids. It was first used by German chemist Adolph Strecker in 1850 to produce α–amino acids.



(i) Name the type of reaction in step **I**. [1]

(ii) Suggest the reagent and condition for step **III**. [1]

(iii) Suggest why a low temperature condition is used in step **II**. [1]

(iv) Step **II** proceeds via 2 stages:

1. acid–base reaction between N in the imine and HCN
2. followed by a nucleophilic attack on C by CN^-

Outline a mechanism for step **II**, showing all charges and using curly arrows to represent the movement of electron pairs. [2]

(v) Suggest why the amino acid synthesised by *Strecker's* method shows no optical activity while that of a naturally occurring amino acid rotates plane polarised light. [2]

[Total: 20]

End of Paper 3